A COUNTEREXAMPLE TO GENERIC INJECTIVITY OF THE TANGENT GAUSS MAP

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We work over $\mathbb{C}$. For a curve $C \subset A$ in an abelian variety, the tangent Gauss map sends a smooth point to its tangent line, translated to the origin:

$$\phi_C : C^{\mathrm{sm}} \longrightarrow \mathbb{P}(T_0 A), \qquad p \longmapsto d(t_{-p})_p(T_p C).$$

We call C nondegenerate if it is not contained in a translate of a proper abelian subvariety.

Theorem 1. *There exists a nondegenerate curve in an abelian threefold whose tangent Gauss map has generic degree* 3.

1. Strategy

Let C be a smooth non-hyperelliptic curve of genus 4, and let $\mathcal{L}$ be a line bundle of degree 3 with $h^0(\mathcal{L}) = 2$. It is base-point-free, since a base point would give a degree-two map to $\mathbb{P}^1$. Thus $|\mathcal{L}| : C \longrightarrow \mathbb{P}^1$ has degree 3. The linear system $\left(\mathcal{L}^{\otimes 2}, \operatorname{Sym}^2 \mathrm{H}^0(\mathcal{L})\right)$ defines the map

$$\phi : C \xrightarrow{|\mathcal{L}|} \mathbb{P}^1 \xrightarrow{\nu_2} \mathbb{P}^2,$$

where ν_2 is the quadratic Veronese embedding. In particular, ϕ has generic degree 3 onto its image and is not generically injective.

To obtain a counterexample by this construction, we seek a pair $(C, \mathcal{L})$ for which this map is a tangent Gauss map. It suffices to impose two further conditions:

(1) $\mathcal{L}^{\otimes 2} \cong \omega_C$;
(2) $\operatorname{Sym}^2 \mathrm{H}^0(\mathcal{L}) \subset \mathrm{H}^0(\omega_C)$ is the $(1,0)$-part of a rational Hodge substructure of $\mathrm{H}^1(C;\mathbb{Q})$.

The second condition allows these three canonical forms to be pulled back from invariant forms on an abelian threefold; we make this passage in Section 2.4.

If we temporarily drop nonhyperellipticity, both conditions can be met. Take a smooth hyperelliptic curve C_0 of genus 4 admitting a tangential cover $\pi_0 : (C_0, p_0) \longrightarrow (E_0, 0)$ at a Weierstrass point p_0. Such examples are supplied by the Lamé spectral curves [3, Table 3 and Theorem L] and [2, Proposition 3.1]. Tangentiality means that $\pi_0^* E_0 \subset \mathrm{Jac}(C_0)$ is tangent to the Abel–Jacobi image at p_0. Put $\mathcal{L}_0 = \mathcal{O}_{C_0}(3p_0)$. Then

$$\mathcal{L}_0^{\otimes 2} \cong \omega_{C_0}, \qquad \operatorname{Sym}^2 \mathrm{H}^0(\mathcal{L}_0) = \mathrm{H}^0\left(\omega_{C_0}(-2p_0)\right).$$

Indeed, a hyperelliptic coordinate x with a double pole at p_0 gives the basis $1, x$ of $\mathrm{H}^0(\mathcal{L}_0)$, with products $1, x, x^2$. Tangentiality gives the Hodge condition, as explained in Lemma 2.1. This seed satisfies the two conditions, but $\mathcal{L}_0$ has a base point at p_0. We therefore deform the pair while preserving both conditions and leaving the hyperelliptic locus.

2. A dimension count for deformations

We first preserve the spin condition and the two sections, then impose compatibility of two different deformations of the cohomology subspaces. A dimension comparison will allow us to remove the hyperelliptic locus.

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2.1. **The deformation space and the spin condition.** Let B be a small simply connected marked deformation space of the spin curve $(C_0, \mathcal{L}_0)$. Thus $\mathcal{L}_t^{\otimes 2} \cong \omega_{C_t}$ throughout B, and $\dim B = 3g - 3 = 9$. To retain two sections, restrict to the theta-null locus

$$S = \left\{ t \in B \;\middle|\; h^0(\mathcal{L}_t) \geqslant 2 \right\}$$

locally defined by one theta constant. Hence $\dim S \geqslant 8$. After shrinking B, upper semicontinuity gives $h^0(\mathcal{L}_t) = 2$ on S.

2.2. **Compatibility of the two deformations.** At the seed, set

$$V_0 := \left(\pi_0^* \,\mathrm{H}^1(E_0; \mathbb{Q})\right)^{\perp} \subset \mathrm{H}^1(C_0; \mathbb{Q}),$$
$$W_0 := \mathrm{Sym}^2 \,\mathrm{H}^0(\mathcal{L}_0) \subset \mathrm{H}^0(\omega_{C_0}),$$

where $\perp$ is taken with respect to the cup-product pairing. Tangentiality gives $V_{0,\mathbb{C}} = W_0 \oplus \overline{W}_0$. Over S, we deform these subspaces in two different ways:

$$V_0 \longmapsto V_t \subset \mathrm{H}^1(C_t; \mathbb{Q}) \qquad \text{by Gauss–Manin parallel transport,}$$
$$W_0 \longmapsto W_t := \mathrm{Sym}^2 \,\mathrm{H}^0(\mathcal{L}_t) \quad \text{by deforming the line bundle.}$$

Thus V_t is a flat rational subspace of dimension 6, whereas $W_t \subset \mathrm{H}^0(\omega_{C_t})$ is a holomorphic subbundle of rank 3. These constructions need not remain compatible: V_t may cease to be a Hodge substructure, and W_t need not lie in its complexification. Their compatibility is precisely the condition we impose:

$$\begin{aligned}
&\begin{cases} V_t \text{ remains a Hodge substructure,} \\ V_{t,\mathbb{C}} \cap \mathrm{H}^0(\omega_{C_t}) = W_t \end{cases} \\
\iff\ & V_{t,\mathbb{C}} = W_t \oplus \overline{W}_t \\
\iff\ & W_t \subseteq V_{t,\mathbb{C}} \\
\iff\ & \mathrm{rk}\left[W_t \longrightarrow \mathrm{H}^1(C_t; \mathbb{C})/V_{t,\mathbb{C}}\right] = 0.
\end{aligned}$$

The middle equivalence follows from rationality of V_t and Hodge type, since $\dim W_t = 3$ and $\dim V_t = 6$. The last condition is the vanishing of a bundle map of ranks 3 and 2, so it imposes at most six equations on S. Consequently, every irreducible component Z of its zero locus through the seed satisfies

$$\dim Z \geqslant 9 - 1 - 6 = 2.$$

Fix one such component Z.

2.3. **Removing the hyperelliptic locus.** To find a nonhyperelliptic point, it remains to bound $\dim(Z \cap \mathcal{H}_4)$, where $\mathcal{H}_4 \subset B$ is the hyperelliptic locus. The following lemma reduces this to counting tangential covers.

Lemma 2.1. *Let C be a smooth hyperelliptic curve of genus 4, and let $\mathcal{L}^{\otimes 2} \cong \omega_C$ with $h^0(\mathcal{L}) = 2$.*

1. *There is a Weierstrass point p such that $\mathcal{L} \cong \mathcal{O}_C(3p)$.*
2. *For this point p, the subspace $\mathrm{Sym}^2 \,\mathrm{H}^0(\mathcal{L}) = \mathrm{H}^0\left(\omega_C(-2p)\right)$ is the $(1,0)$-part of a rational Hodge substructure if and only if C admits a tangential cover at p.*

Proof.

1. Fix a Weierstrass point w. If $\mathcal{L}$ were base-point-free, its degree-three map and the degree-two map $|\mathcal{O}_C(2w)|$ would give a birational map to a curve of bidegree $(2,3)$ in $\mathbb{P}^1 \times \mathbb{P}^1$, of arithmetic genus 2. Thus $\mathcal{L}$ has a base point p, and uniqueness of the degree-two linear system gives $\mathcal{L}(-p) \cong \mathcal{O}_C(2w)$. Using $\omega_C \cong \mathcal{O}_C(6w)$, the spin condition gives $\mathcal{O}_C(2p) \cong \mathcal{O}_C(2w)$. Hence p is Weierstrass and $\mathcal{L} \cong \mathcal{O}_C(3p)$.
2. Under $T_0^* \,\mathrm{Jac}(C) = \mathrm{H}^0(\omega_C)$, the space $\mathrm{H}^0\left(\omega_C(-2p)\right) = \mathrm{H}^0\left(\omega_C(-p)\right)$ is the annihilator of the tangent line at p. Its being the $(1,0)$-part of a rational Hodge substructure is therefore equivalent to this line being the tangent space of an elliptic subvariety of $\mathrm{Jac}(C)$. Dualizing the inclusion and composing with the Abel–Jacobi map gives a tangential cover; the converse follows from the same annihilator description.

□

Consequently, the hyperelliptic points of Z give tangential covers $\pi_t : (C_t, p_t) \longrightarrow (E_t, 0)$. The complementary Hodge substructure $V_t^\perp$ has rank 2 and determines E_t up to isogeny. Choosing its saturated integral lattice fixes the pullback map on integral cohomology under local markings. The cup-product pairing then shows that $\deg \pi_t$ is locally constant.

Remark 2.1. *For a fixed elliptic curve and degree there are only finitely many hyperelliptic tangential covers* [4, Proposition 2.5, 3.; Section 2, Remark 1, 3.–4.]. *Indeed, the relation* $2n+1 = \sum_{i=0}^{3} \mu_i^2 + 4d$ *leaves finitely many types for a fixed degree* n*, and each corresponding Severi set is finite.*

Since elliptic curves have one modulus and the choices of Weierstrass points p_t are finite,

$$\dim (Z \cap \mathcal{H}_4) \leqslant 1 < 2 \leqslant \dim Z.$$

We can therefore choose a nonhyperelliptic point of Z.

2.4. **Passing to a curve in an abelian threefold.**

Conclusion of the proof of Theorem 1.

(1) *Constructing the map.* For the chosen point, the rational Hodge substructure V gives a quotient $q : \mathrm{Jac}(C) \longrightarrow A$ with connected kernel and $q^* \mathrm{H}^0(\Omega^1_A) = \mathrm{Sym}^2 \mathrm{H}^0(\mathcal{L})$. Compose the Abel–Jacobi map based at $b \in C$ with q: $f = q \circ j_b : C \longrightarrow A$. The pulled-back invariant forms are the products of sections of $\mathcal{L}$, so f is immersive and its tangent map is ϕ. Its image contains the origin and generates A, hence is nondegenerate.

(2) *The normalization and the Gauss map.* If f had degree $e > 1$ onto the normalization of its image, then $e \mid 3$, since ϕ factors through that normalization. Thus $e = 3$, and the normalization would map with degree 1 to the conic $\nu_2(\mathbb{P}^1)$. This is impossible in an abelian variety. Therefore C is the normalization of $f(C)$ and the tangent Gauss map of $f(C)$ has generic degree 3.

□

3. Simplicity and symmetries

Theorem 2.

(1) *The abelian threefold in Theorem 1 can be chosen simple.*

(2) *For this choice, the resulting curve is preserved by no nonidentity automorphism of* A *with scalar differential.*

Proof.

(1) **A seed with simple quotient.** Take $E_0 : y^2 = 4x^3 - 4x - 1$, let e_1, e_2, e_3 be the roots of $4e^3 - 4e - 1$, and take the smooth projective model of

$$C_0 : \quad \nu^2 = (B^3 - 208B + 560) \prod_{i=1}^{3} \left(B^2 + 10e_i B - 35e_i^2 - 28\right).$$

This is the genus-four Lamé spectral curve, with a degree-ten cover of E_0 tangential at infinity [3, Table 3 and Theorem L] and [2, Proposition 3.1 and the discussion after (17)]. The cover is defined over $\mathbb{Q}$, and E_0 has discriminant 37. Modulo 11, the polynomial on the right is the square-free polynomial $B^9 + B^4 + 6B^3 + 9B^2 + 7B + 5$; hence C_0 has genus 4 and good reduction. Direct counting gives

$$(\#C_0(\mathbb{F}_{11^r}))_{r=1}^{4} = (10, 120, 1312, 15312), \qquad \#E_0(\mathbb{F}_{11}) = 7.$$

Newton's identities and division by the elliptic factor $T^2 - 5T + 11$ give the Frobenius polynomial of $A_0 = \mathrm{Jac}(C_0)/\pi_0^* E_0$:

$$h(T) = T^6 + 3T^5 + 5T^4 - 14T^3 + 55T^2 + 363T + 1331.$$

It is irreducible modulo 3, and $11 \nmid 14$, so this reduction is simple and ordinary. An ordinary simple abelian threefold over a finite field of odd cardinality is absolutely simple unless its

Frobenius polynomial has the form $T^6 + cT^3 + q^3$ [1, Proposition 3.4]. Thus this reduction is absolutely simple, and so is A_0 in characteristic zero, since a proper abelian subvariety would specialize to one after a finite extension.

Preserving simplicity under deformation. Choose this seed in the construction of Z. For every nonzero proper rational subspace $R \subset V_0$ on which the polarization is nondegenerate, let e_R be the orthogonal projector. The condition

$$e_R(W_t) \subset W_t$$

defines a closed analytic subset of Z. It is proper because it misses the seed, and it holds precisely when R is a Hodge substructure. There are only countably many R, so Baire's theorem allows us to choose a point outside these subsets and $\mathcal{H}_4$. Its Hodge structure V_t is simple, hence the corresponding abelian threefold A is simple.

(2) **Reducing to an automorphism of order three.** Suppose $\tau \in \mathrm{Aut}(A)$ preserves $f(C)$ and has scalar differential, using translations to identify tangent spaces. It lifts to an automorphism σ of C with $\phi \circ \sigma = \phi$. If σ is the identity, then τ fixes a generating curve pointwise and is the identity. Otherwise, the order of σ divides $\deg \phi = 3$, so it is 3 and $C/\langle\sigma\rangle \cong \mathbb{P}^1$. Riemann–Hurwitz gives a fixed point of σ.

Contradicting simplicity. Taking the image of this fixed point as the origin makes τ a group automorphism u of order 3. Its scalar differential is a primitive cube root of unity λ. Then $\mathbb{Q}(\lambda)$ acts on $\mathrm{H}_1(A;\mathbb{Q})$, and the complex structure on real homology is $(u_* - \mathrm{Re}(\lambda)) \,/\, \mathrm{Im}(\lambda)$. Every one-dimensional $\mathbb{Q}(\lambda)$-subspace therefore has a complex-invariant realification and defines an elliptic subvariety of A, contradicting simplicity.

□

References

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